

Aarsh Desai

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in LinkedIn

GitHub

Portfolio

Experience

Nanonets

January 2025 - Present

Deep Learning Engineer · July 2025 - Present

Bengaluru, Karnataka

- Designed a multi-layer memory framework for data extraction agents that turns real-time user feedback and historical data patterns into rules by measuring user intent. It flags conflicting instructions and lets agents adapt as business logic and edge cases change.
- Improved domain specific multimodal pipelines for structured data extraction from documents by focusing on reducing latency, cost and hallucinations on noisy real-world inputs.
- Led the precision work for enterprise customers, diagnosed why certain methods kept failing and shipped targeted fixes to match the strict accuracy SLA requirements across different workflows.

Deep Learning Intern · January 2025 - July 2025

- Fine-tuned SigLIP and Qwen2.5-VL encoders using contrastive learning to produce embeddings for clustering and retrieval. The final model outperformed SOTA baselines by 30% on highly unstructured layouts and templates.
- Built a human-in-the-loop annotation framework that combined active learning with custom labelling UIs and made high quality label collection 70% faster.

Robert Bosch Centre for Data Science and AI, IIT Madras

June 2024 - November 2024

Research Intern

Remote

- Worked on a multimodal stock prediction system for swing trading that used transformer-based sentiment from social media and financial news along with technical indicators.
- Tuned the time-series forecasting models and feature selection, which improved prediction accuracy by 25% over historical price baselines.

Centre for Educational Technology, IIT Bombay

May 2023 - September 2024

Research Intern

Mumbai, Maharashtra

- Built a speaker diarization pipeline using PyAnnote and Whisper for Indo-English classroom conversations (140+ hours). It outperformed existing SOTA models by 65% and won Best Student Paper at IEEE TALE 2024.
- Worked on AffectBots, a web-based tutoring system that recognises student emotion from video and adjusts the teaching methods accordingly.

Vocab.Ai

August 2023 - August 2024

MLOps Intern

Remote

- Built a pipeline for ingesting customer service chats in real time through Google Business APIs, and a dashboard showing how CS agent behaviour affects TAT and CSAT scores.

Dept of CSE, NIT Puducherry

December 2023 - January 2024

Deep Learning Intern

Karaikal, Puducherry

- Cut the energy use of lung nodule classification and segmentation models so they can run on edge devices in low-resource areas. Compared depthwise separable CNNs, pruning and quantization on LIDC-IDRI with minimal accuracy loss.

Education

Indian Institute of Information Technology, Dharwad

May 2025

BTech in Data Science and Artificial Intelligence (Cumulative GPA: 9.19)

Dharwad, Karnataka

G.D. Goenka International School

May 2021

CBSE HSC (Percentage: 94.4)

Surat, Gujarat

Technical Skills

Focus Areas: Vision-Language Models, Agentic Systems, Multimodal Learning

ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, OpenCV

Training & Serving: vLLM, Triton Inference Server, FastAPI, quantization, pruning

Languages & Data: Python, C/C++, Go, SQL, PostgreSQL, MongoDB, Redis, Cassandra, Elasticsearch

Infrastructure: AWS, GCP, Docker, Kubernetes, Jenkins, Argo CD, GitHub Actions, Grafana

Projects

Attention and Emotion Sensing for Online Learning | TensorFlow, OpenCV, dlib, WebGazer.js | 

May 2023

- Built a real-time webcam tool that flags drowsy, distracted or disengaged students for teachers of students with special needs, using facial landmarks to detect and track head pose, eye closure and gaze.
- Trained a CNN from scratch on the FER-2013 dataset and reached 62.7% accuracy across 7 emotions.
- Later brought the same idea to a study platform's PDF viewer with in-browser gaze tracking and an alert when the reader looks away. The entire tracking and alarm logic was offline so user video never left the user's personal device.

Research Publications

- **Desai, A.**, Kartik, N. V. J. K., Gupta, P., M, V., Vanahalli, M. K., Rajendran, R., T S, A., & Biswas, G. (2025). Exploring Emotional Trajectories through Valence and Arousal in Open-Ended Math Learning Environments. *International Conference on Technology 4 Education (T4E 2025)*. NSF Award: 2112635
- Sharma, V., Kartik, N. V. J. K., Vinayak, **Desai, A.**, Gupta, P., Vanahalli, M. K., Prasad, B. M., Bankapur, S. S., & Girish, G. N. (2025). Energy-Efficient Depthwise Separable CNNs for Classification and Segmentation of Lung Cancer Nodules. *IEEE India Council International Subsections Conference (INDISCON 2025)*. DOI: 10.1109/INDISCON66021.2025.11254247
- **Desai, A.**, Kartik, N. V. J. K., Gupta, P., Vinayak, V., T S, A., Vanahalli, M. K., & Rajendran, R. (2024). Advancing Speaker Diarization With Whisper Speech Recognition for Different Learning Environments. *IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE 2024)*. DOI: 10.1109/TALE62452.2024.10834319
- Kartik, N. V. J. K., Gupta, P., M, V., **Desai, A.**, Badhe, V., T S, A., Vanahalli, M.K., & Rajendran, R. (2024). Unlocking the Triggers: Automating the Identification of Triggers of Socially Shared Metacognitive Regulation in Collaborative Problem-Solving. *International Conference on Technology 4 Education (T4E 2024)*. DOI: 10.1007/978-981-95-1734-3_6

Achievements

- **Director's Gold Medal** for the best all-round graduating student in the batch of 2025 at IIIT Dharwad.
- **Best Student Paper Award** at IEEE TALE 2024.
- **1st place** at the Ganglia Tech Collegiate Hackathon.
- **Global Rank 19** in CodeChef Starters 57, peak 4-star rating on CodeChef and top 3% on LeetCode.

Positions of Responsibility

President Data Science and AI Society, IIIT Dharwad	October 2023 - October 2024
Club Lead InQuizitive, the Quiz Club of IIIT Dharwad	October 2022 - December 2023